

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC406 - ENGINEERING GEOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

1 The core of the Earth is mainly composed of:

- a. Silica and magnesium
- b. Iron and nickel
- c. Oxygen and nitroge
- d. Aluminum and copper

2 Meanders are formed due to:

- a. River erosion
- b. Wind deposition
- c. Mountain uplift
- d. Earthquakes

3 Which mineral property is tested using Mohs scale?

- a. Luster
- b. Hardness
- c. Cleavage
- d. Streak

4 Marble is formed from the metamorphism of:

- a. Shale
- b. Limestone
- c. Sandstone
- d. Granite

5 Which instrument is commonly used to measure strike and dip?

- a. Compass Clinometer
- b. Seismograph
- c. Plumb bob
- d. Level instrument

- 6 Strike-slip faults are characterized by:
- Horizontal movement
 - Vertical movement
 - Overturning folds
 - Unconformity surface
- 7 Core recovery is expressed as:
- Percentage of core length recovered
 - Rock strength
 - Volume of water
 - Drill bit speed
- 8 Preliminary geological investigations help in:
- Planning further detailed studies
 - Avoiding all surveys
 - Eliminating site visits
 - Guaranteeing project success
- 9 Exploratory drilling provides:
- Direct information of subsurface
 - Remote sensing data
 - Geophysical survey only
 - Aerial photographs only
- 10 Dams on carbonate rocks face issues like:
- Solution cavities
 - High strength
 - No permeability
 - Flat bedding
- 11 Folded strata in dam sites may cause:
- Leakage along folds
 - Perfect stability
 - No fractures
 - Zero seepage
- 12 Grouting is used for:
- Reducing seepage
 - Painting dam walls
 - Increasing porosity
 - Erosion control

Q2. Solve the following.

- What is the difference between convergent and divergent plate boundaries? 6
- Give a detailed account of types of mountains and their origin with suitable examples. 6

Q3. Solve the following.

- Give a detailed account of the physical properties of minerals useful for the identification of minerals. 6
- Write a detailed classification of sedimentary rocks based on their textures and structures of sedimentary rocks. 6

Q4. Solve Any Two of the following.

- A) Define “unconformity” and describe various types of unconformities. 6
- B) What are the selection criteria for the selection of stones for construction? 6
- C) Explain various factors behind groundwater level fluctuations. 6

Q5. Solve Any Two of the following.

- A) Write a detailed note on core preservation and arrangement methods. 6
- B) What are the advantages and limitations of the rock quality designation technique? 6
- C) Explain the complete procedure of engineering geological investigation. 6

Q6. Solve Any Two of the following.

- A) Define a bridge and describe various types of bridges with neat labeled diagrams. 6
- B) Write a detailed note on the suitable and unsuitable characteristics of igneous rocks at tunnels. 6
- C) Explain the various favourable and unfavourable geological conditions for Dam site. 6

*** End ***